Legal eBilling and Billing Operations Analysis

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Executive summary

I used Excel, SQL Server and Power BI to analyse a synthetic law firm billing dataset. I chose this topic because it relates to my eBilling work experience. The aim was to understand billing activity, rejections, reductions and appeal recoveries, and compare invoicing with recorded matter budgets.

After cleaning, the dataset contains 1,985 invoices, 150 matters and 15 clients. Gross invoicing totals $50,602,194.65. Reductions total $689,140.44, of which $339,087.46 is offset by accepted appeals. This leaves $350,052.98 in reductions after appeals.

Rate approval and budget approval are the two main rejection reasons. Together they account for 264 of 340 rejected invoices, or 77.65%. These would be the first areas I would review before invoice submission.

The dashboard shows budget utilisation of 137.04% and 126 active matters. Net Accepted Amount is $33,109,794.18. This includes Approved, Exported for Payment, Paid and Reduced invoices after reductions and accepted appeals. It represents accepted billing under the project definition, not cash received.

All data is AI-generated. The findings describe this practice dataset, and the recommendations have not been tested in a real business.

Methodology

I kept the raw workbook separately and cleaned a copy in Excel. I checked duplicates, missing values, text consistency, negative amounts and date order. I then calculated descriptive statistics to understand the amounts and their spread.

In SQL Server, I used decimal financial columns and wrote queries for budgets, invoice statuses, reduction and rejection reasons, client ranking, monthly billing, processing time and high-value invoices. I compared the invoice count and financial totals with Excel.

In Power BI, I linked Matters_Budgets to Invoices_Log through Matter Number using a one-to-many relationship with single-direction filtering. I created the measures and tested the client, practice area, portal and date slicers. The cleaned workbook and Power BI source were also compared by their business keys.

The rate reference table contains 100 records. Only 37 invoices matched a rate record by timekeeper name, client and portal, so I excluded hourly-rate analysis from the SQL script and Power BI dashboard. The rate table is included for reference.

Data cleaning and checks

I kept the original workbook in Data/Raw_Dataset.xlsx and saved the corrected version as Data/Cleaned_Dataset.xlsx. The steps below cover the initial cleaning and the fixes made during review.

1. Removed 15 exact duplicate invoice rows from the original 2,000 rows. The final table has 1,985 unique Invoice IDs.

2. Standardised casing and surrounding spaces. This changed 282 invoice-status values, 796 portal values and 341 biller values. Portal text was also standardised in 40 rate-reference rows.

3. Replaced 1,699 non-applicable blanks with zero in each of the reduction and accepted-appeal columns. During review, I found that 242 positive appeal values had also been accidentally changed to zero. I restored these by Invoice ID, adding back $302,600.09.

4. Corrected five injected negative invoice signs using ABS. The affected IDs are INV-00249, INV-00519, INV-00748, INV-01406 and INV-01653. This treatment follows the synthetic-data design; an actual credit would need a different review.

5. Reordered the three event dates on eight invoices using MIN, MEDIAN and MAX. These are assumed corrections, not confirmed event dates. All eight remain in the analysis. Their original and adjusted dates are listed in the appendix.

6. Restored unintended date edits on seven other invoices: INV-01343, INV-00456, INV-01213, INV-01001, INV-00234, INV-01381 and INV-00137. This restored 25 date fields. All Status Date values now match the raw source.

7. Retained the four injected invoices above $250,000. Together they total $1,195,563.94, or 2.36% of gross invoicing. They are synthetic examples, not verified real bills.

8. Standardized timekeeper IDs by name, updating 17 rows. The rate table has 100 rows and 83 distinct timekeeper IDs because one timekeeper can have multiple rate records. The final billing calculations exclude these rates.

Import corrections

An early Power Query step converted accepted appeals to Whole Number and rounded away cents. Changing that original step to Fixed Decimal Number corrected a $7.54 difference in Net Accepted Amount. Changing only the display format had not restored the missing cents.

The first SQL import showed cent differences on three large invoices. I reimported the invoice data with decimal(18,10), then changed the three financial columns to decimal(18,2). The four large invoices and the overall totals then matched Excel. The exact earlier import stage that caused the differences was not established.

Final checks

The final checks found 1,985 unique Invoice IDs and 150 unique matter records. Every invoice links to a matter, and the client names agree between those tables. Accepted appeals do not exceed reductions, and reductions do not exceed invoice amounts. Gross invoicing, reductions and accepted appeals match across the corrected Excel data, SQL and Power BI.

Descriptive statistics and interpretation

Amounts below are USD. I used population standard deviation in Excel to describe the supplied dataset. The first table includes all invoices, including zero reductions and appeals.

Statistic	Invoice amount	Reduction	Accepted appeal
Count	1,985	1,985	1,985
Mean	25,492.29	347.17	170.82
Median	25,545.62	0.00	0.00
Standard deviation	18,854.23	1,097.48	612.65
Minimum	587.63	0.00	0.00
Maximum	337,332.80	7,763.84	6,875.50

The mean and median invoice are both around $25,500. However, the standard deviation and maximum show that invoice values vary widely. Similar mean and median values do not rule out outliers. Excluding the four invoices above $250,000 lowers the average to $24,940.25; the final analysis retains them.

The median reduction and appeal are zero because most invoices have neither amount. Looking only at the 286 Reduced invoices, the average reduction is $2,409.58 and the average accepted appeal is $1,185.62. The appeal average includes three Reduced invoices with zero accepted appeal.

Statistic	Matter budget	Accrual amount
Count	150	150
Mean	246,158.59	31,321.07
Median	245,692.00	29,231.63
Standard deviation	133,737.89	23,164.24
Minimum	30,716.00	101.30
Maximum	499,753.00	84,628.87

Matter budgets differ considerably, so an overall average does not describe every matter. Accruals represent estimated unbilled work. There is no accrual period in the source, so I did not add these amounts to invoicing to calculate total exposure.

The 100 reference-rate records have a mean of $412.37, median of $392.00 and standard deviation of $165.88. Rates range from $156 to $795. These statistics describe the reference file only, because its invoice matching is too limited to describe rates across the portfolio.

Findings and recommendations

Approval issues account for most rejected invoices. Rate Approval Required covers 147 invoices worth $3,830,138.06; Budget Not Approved covers 117 worth $2,865,471.32. Together they represent 77.65% of rejected invoice count and 78.36% of rejected value. I would check these approvals before submission and maintain a follow-up list for affected invoices. Their face value is not guaranteed future recovery.

Excessive hours is the largest reduction reason at $132,984.00, or 19.30% of reductions. Billing guideline adjustments contribute $128,918.22 and administrative time $112,614.02. Together these reasons make up 54.35%. Reviewing hours and narratives against client guidelines would be a practical next step.

Accepted appeals offset 49.20% of reduction value. The remaining $350,052.98 could be reviewed for appeal eligibility and supporting evidence. It is not necessarily a final write-off. This percentage is not an appeal success rate because submitted appeal values are missing.

Summit Retail Holdings has the largest gross billing value at $4,296,147.75, or 8.49%. Redwood Manufacturing and Westbridge Hospitality follow. The top three account for 23.62% of gross invoicing. Legal Tracker and CounselLink together handle 61.01% of invoice count. This can help identify where most client and portal activity sits.

Gross invoicing is 137.04% of recorded budgets, with 98 of 150 matters above their recorded allocation. Before treating this as confirmed overspending, I would check the budget period, version and approval. The source includes 92 Approved, 35 Pending Approval and 23 Not Approved budgets.

For all 464 Paid invoices, the average interval from submission to Status Date is 38.08 days. Three have assumed date adjustments. This measures time to the recorded Paid-status update; it is not a verified cash-receipt measure.

Dashboard definitions

Net Accepted Amount includes Approved, Exported for Payment, Paid and Reduced invoices only. For those invoices, it is invoice amount minus reduction plus accepted appeal. Total Reduction Amount is the gross reduction before appeals. Reduction Rate is gross reductions divided by gross invoicing, or 1.36% overall.

Budget Utilization is gross invoicing divided by the sum of recorded matter budgets, including all approval statuses. Client and practice filters affect both. Date and portal filters affect invoices while budgets stay fixed because the source has no monthly or portal budget allocation. A card is used so utilization above 100% is visible.

Active Matters includes Open and On Hold matters: 105 + 21 = 126. The dashboard also shows client invoicing against budgets, invoice count by portal and a detail table by client, matter and portal.

Limitations

This is synthetic data with deliberately designed patterns. Each invoice has one recorded status, so the 17.13% currently marked Rejected is not a first-pass rejection rate. Invoice dates cover 1 January 2025 to 15 October 2026, making October partial. Status dates extend to 24 December 2026 and are treated as simulated outcomes rather than a historical position.

Receipt amounts, verified payment dates, due dates and outstanding balances are missing. The project therefore does not calculate DSO, overdue ageing or verified collections. The assumed date changes and missing budget periods also limit the processing-time and budget conclusions.

Appendix of date adjustments

I sorted the three existing dates separately for each of the eight affected invoices. I used MIN for Invoice Date, MEDIAN for Date Assigned and MAX for Date Submitted. This gives each row a sensible sequence based on the assumption that its three dates were entered in the wrong order.

All eight invoices remain in the analysis. The table shows each invoice's original and adjusted dates in the order Invoice Date, Date Assigned and Date Submitted. The assigned roles for those dates are assumptions.

Invoice ID	Original dates	Adjusted dates
INV-00041	2026-09-14 2026-09-14 2026-09-11	2026-09-11 2026-09-14 2026-09-14
INV-00855	2026-08-30 2026-09-06 2026-09-01	2026-08-30 2026-09-01 2026-09-06
INV-01111	2026-02-18 2026-02-22 2026-02-21	2026-02-18 2026-02-21 2026-02-22
INV-01338	2025-12-07 2025-12-10 2025-12-07	2025-12-07 2025-12-07 2025-12-10
INV-01399	2026-02-26 2026-03-01 2026-02-25	2026-02-25 2026-02-26 2026-03-01
INV-01510	2026-04-11 2026-04-13 2026-04-12	2026-04-11 2026-04-12 2026-04-13
INV-01581	2025-04-16 2025-04-19 2025-04-14	2025-04-14 2025-04-16 2025-04-19
INV-01887	2025-07-28 2025-07-28 2025-07-27	2025-07-27 2025-07-28 2025-07-28